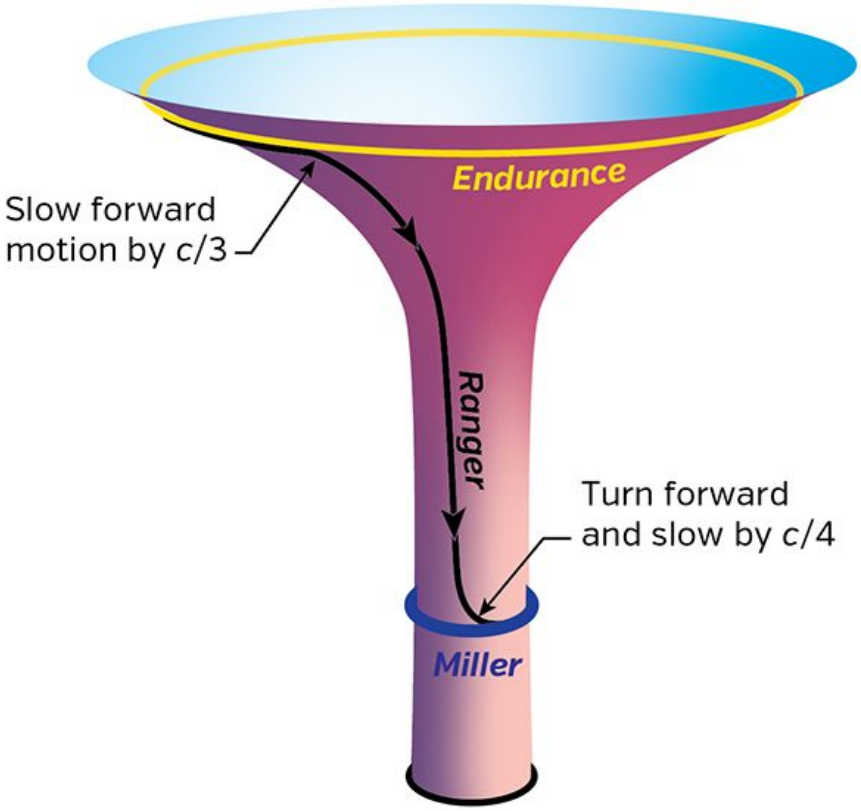


Fig. 7.1. The Ranger's trip to Miller's planet, in my interpretation of *Interstellar*.

What mechanism can Cooper, the Ranger's pilot, possibly use to produce these huge velocity changes?

Twenty-First-Century Technology

The required changes of velocity, roughly $c/3$, are 100,000 kilometers per second (per *second*, not per hour!).

By contrast, the most powerful rockets we humans have today can reach 15 kilometers per second: seven thousand times too slow. In *Interstellar*, the *Endurance* travels from Earth to Saturn in two years at an average speed of 20 kilometers per second, five thousand times too slow. The fastest that human spacecraft are likely to achieve in the twenty-first century, I think, is 300 kilometers per second. That would